

Test Report -- DL-06 Persistence, EMA Learning, APScheduler, Stats

Branch: feat/dl06-persistence-learning Date: 2026-05-31 Tester: automated via PowerShell + curl against live server (uvicorn, port 8000)

What was tested

All four DL-06 pillars: 1. SQLite persistence -- agent_log and approvals survive server restarts 2. EMA supplier score updates -- on-time delivery raises score, late delivery lowers it 3. APScheduler -- scheduler starts with server, next_run visible in /api/stats 4. Stats endpoint + betsy.html panel -- correct shape and values

Test results

#	Test	Expected	Result	Pass
T1	Server health + scheduler active	scheduler_active: true, next_run set	Both present	?
T2	POST to /api/agent-log writes to SQLite	Row appears in betsy.db agent_log table	id=7, tr	
T3	POST to /api/approvals writes to SQLite	Row appears in betsy.db approvals table	decision	
T4	Server restart -- log + approvals reload from DB	7 log entries + 1 pending approval present		
T5a	EMA -- on-time delivery raises score	$0.2 \times 1.0 + 0.8 \times 0.97 = 0.976$	Score 0.9700	-> 0.97
T5b	EMA -- 5-day late delivery lowers score	$0.2 \times 0.5 + 0.8 \times 0.976 = 0.8808$	Score 0.9760	->
T6	GET /api/stats returns correct shape	All fields present, scheduler info populated	decisi	
T7	Approve queued item -- DB row updated	status=approved, resolved_at set, queue empties	st	
T8	Double-resolve guard	400	HTTP 400	?

9/9 passed (including 2/2 EMA formula checks).

EMA formula verification

Supplier: PrecisionParts GmbH (SUP-004), baseline score 0.97.

Event	Lateness	Performance	Formula	Before	After	Match
On-time delivery	0 days	1.0	$0.2 \times 1.0 + 0.8 \times 0.9700$	0.9700	0.9760	?
5-day late delivery	5 days	0.5	$0.2 \times 0.5 + 0.8 \times 0.9760$	0.9760	0.8808	?

Performance decay: $\max(0, 1.0 - \text{lateness} \times 0.1)$ -- 10 days late = 0.0 performance.

Persistence verified

State before restart: - 7 agent_log entries (including EMA score_updated events and test_persistence entry) - 1 pending approval (test-dl06-persist-001, generate_po, ?5,500)

State after server restart -- reloaded from betsy.db: - 7 agent_log entries ? - 1 pending approval ?

No data lost. betsy.db is created at project root on first server start.

APScheduler confirmed

- scheduler_active: true visible in /api/stats immediately after server start
 - next_run field shows ISO timestamp of the next scheduled pipeline run
 - Default interval: 30 minutes (override with AGENT_INTERVAL_MINUTES env var)
 - Set AGENT_INTERVAL_MINUTES=1 for testing -- auto-run fires within 60s
-

Stats panel (betsy.html)

GET /api/stats returns:

```
{
  "decisions_total": 9,
  "decisions_auto": 0,
  "decisions_human": 0,
  "ema_updates": 6,
  "auto_rate_pct": 0.0,
  "pending_approvals": 1,
  "queue_value_eur": 5500.0,
  "last_run": null,
  "scheduler_active": true,
  "next_run": "2026-05-31T18:44:57..."
}
```

betsy.html stats panel (5 cards): decisions made · autonomous rate · awaiting your OK · value in queue · next auto-run. Updates every 5 seconds with the rest of the dashboard.

Known limitations (documented in DL-06)

- Supplier scores are session-persistent only -- EMA updates apply to in-memory state loaded from mock_data/suppliers.json. A server restart reloads baseline scores. Score learning is demonstrated within a session; persisting scores to the DB is a future step.
- Concurrent runs -- if a scheduled run and a manual run overlap, both can enqueue approvals simultaneously. No locking between pipeline threads.
- Score drift on scenario reset -- state.reset() reloads suppliers from the original JSON, undoing any EMA changes made during the session. Don't mix EMA tests with scenario injection in the same flow.